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TREND-BASED MOCK PAPER

GATE Mathematics

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Mathematics
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section B: Mathematics

Q1. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q2. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q3. [1-NAT] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-NAT] [1 mark]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q5. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q6. [1-MCQ] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q7. [1-NAT] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q8. [1-NAT] [1 mark]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q10. [1-MCQ] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q11. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q12. [1-MCQ] [1 mark]

Which of the following is a probability generating function for $\operatorname{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q13. [1-NAT] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q14. [1-MCQ] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| \cdot \|v\|$

- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q15. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) ∞

Q16. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q17. [1-NAT] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q19. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q20. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q21. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q22. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q23. [2-MCQ] [2 marks]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q24. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q25. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q26. [2-MSQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q27. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)} \cdot u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q28. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q29. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MSQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q31. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MSQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
 - (B) 8
 - (C) 16
 - (D) 12
-

Q33. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
 - (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
 - (C) $[(3+2\sqrt{2})^{50}, \dots]$
 - (D) Cannot determine
-

Q34. [2-MCQ] [2 marks]

The Laplace transform of $t \cdot e^{-2t}$ is:

- (A) $1/(s+2)^2$
 - (B) $1/(s-2)^2$
 - (C) $s/(s+2)^2$
 - (D) $2/(s+2)^2$
-

Q35. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
 - (B) $1/(1-z)$
 - (C) $\ln(1-z)$
 - (D) $e^{-\lambda}(1+z)$
-

Q36. [1-NAT] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

(Numerical Answer Type — Enter the numerical value)

Q37. [1-MCQ] [1 mark]

Sum of first n odd numbers:

- (A) n^2

- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q38. [1-MCQ] [1 mark]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q39. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q40. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q41. [2-NAT] [2 marks]

Compound interest: P at r% for 2 years =

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q43. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q44. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q45. [2-NAT] [2 marks]

Compound interest: P at r% for 2 years =

(Numerical Answer Type — Enter the numerical value)

Q46. [2-NAT] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

(Numerical Answer Type — Enter the numerical value)

Q47. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
 - (B) $2n$
 - (C) $n(n+1)$
 - (D) $n(n-1)$
-

Q48. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
 - (B) $2n$
 - (C) $n(n+1)$
 - (D) $n(n-1)$
-

Q49. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
 - (B) $P(1+2r/100)$
 - (C) $P + Pr/100 \times 2$
 - (D) $P \times r/100$
-

Q50. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
 - (B) 27
 - (C) 9
 - (D) 3
-

Q51. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
 - (B) $2n$
 - (C) $n(n+1)$
 - (D) $n(n-1)$
-

Q52. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q53. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q54. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q55. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q56. [1-NAT] [1 mark]

Rank of 3x3 identity:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-NAT] [1 mark]

Inverse of AB = ?

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both +, -, ×
- (D) None

Q59. [1-MCQ] [1 mark]

Identity element property:

- (A) $a \times e = e \times a = a$
- (B) $a \times e = e$
- (C) $e \times a = e$
- (D) $a \times e = a$

Q60. [1-MCQ] [1 mark]

Order of $y'' + 3y' + 2y = 0$:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q61. [1-MCQ] [1 mark]

Limit of $\sin(x)/x$ as $x \rightarrow 0$:

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

Q62. [1-MCQ] [1 mark]

Cauchy sequence:

- (A) Converges always
- (B) Not always
- (C) Never
- (D) Random

Q63. [1-NAT] [1 mark]

Union of open sets is:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

Normal distribution PDF:

- (A) $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$
- (B) $\exp(-x)$
- (C) $1/x$
- (D) x^n

Q65. [1-NAT] [1 mark]

$f(x)=|x|$ at $x=0$:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	D	1
2	1NAT	A	1
3	1MCQ	C	1
4	1MCQ	A	1
5	1MCQ	C	1
6	1MCQ	B	1
7	1MCQ	B	1
8	1NAT	A	1
9	1MCQ	A	1
10	1NAT	B	1
11	1MCQ	B	1
12	1NAT	A	1
13	1MCQ	A	1

14	1MCQ	A	1
15	1NAT	A	1
16	1NAT	C	1
17	1NAT	C	1
18	1MCQ	A	1
19	1MCQ	A	1
20	1NAT	A	1
21	1NAT	C	1
22	1MCQ	A	1
23	1MCQ	C	1
24	1MCQ	A	1
25	1MCQ	A	1
26	1NAT	A	1
27	1MCQ	A	1
28	1MCQ	B	1

29	1MCQ	A	1
30	1NAT	B	1
31	2MCQ	A	2
32	2MCQ	A	2
33	2MCQ	B	2
34	2MCQ	A	2
35	2MCQ	B	2
36	2MCQ	A	2
37	2MCQ	B	2
38	2MCQ	C	2
39	2MSQ	A	2
40	2MCQ	A	2
41	2MCQ	A	2
42	2NAT	A	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2MCQ	A	2
46	2MCQ	B	2
47	2MCQ	A	2
48	2NAT	A	2
49	2NAT	A	2
50	2NAT	A	2
51	2MCQ	A	2
52	2MSQ	B	2
53	2NAT	B	2
54	2MCQ	A	2
55	2MCQ	A	2
56	2MSQ	C	2
57	2MCQ	A	2
58	2MCQ	A	2

59	2MCQ	A	2
60	2MCQ	A	2
61	2MCQ	A	2
62	2MCQ	A	2
63	2MCQ	A	2
64	2MCQ	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: D

Solution:

Correct Answer: D

Subject: Algebra | Topic: Linear Algebra | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Algebra | Topic: Matrix | Difficulty: difficult

Q3.

Answer: C

Solution:

Correct Answer: C

Subject: Algebra | Topic: Ring Theory | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Algebra | Topic: Group Theory | Difficulty: easy

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Calculus | Topic: ODE | Difficulty: easy

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Calculus | Topic: Real Analysis | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Analysis | Topic: Metric Spaces | Difficulty: moderate

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Topology | Topic: Open Sets | Difficulty: difficult

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Probability | Topic: Distributions | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Real Analysis | Topic: Continuity | Difficulty: easy

Q11.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q12.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q13.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q14.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q15.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q16.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q17.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees $= n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q18.

Answer: A

Solution:

Area $= \pi*r^2$. Circumference $= 2*\pi*r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q21.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q22.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q25.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q28.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q30.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

$$L\{t^*f(t)\} = -d/ds(F(s)). F(s) = 1/(s+2). -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: C

Solution:

$$y = A*\cos(2x) + B*\sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. x^3 + 1/x^3 = 27 - 3*3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q41.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q42.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q43.

Answer: A

Solution:

$$L\{e^{(-at)}\} = 1/(s+a). \text{ Here } a=3: 1/(s+3).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q44.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q45.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q46.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q48.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q49.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q50.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q51.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q52.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q55.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q56.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q57.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q58.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q59.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q60.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q61.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q62.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q63.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q64.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q65.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 35 questions, 53 marks
Miscellaneous: 20 questions, 37 marks
Algebra: 4 questions, 4 marks
Calculus: 2 questions, 2 marks
Analysis: 1 questions, 1 marks
Topology: 1 questions, 1 marks
Probability: 1 questions, 1 marks
Real Analysis: 1 questions, 1 marks

Difficulty Distribution:

Easy: 17 | Moderate: 34 | Difficult: 14

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.